iLearn Practice Exam

11BR · Chapter / Topic 2

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Topic 2: Communication

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 6 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A company has a LAN (local area network).

(a) Give **two** benefits of connecting computers to a LAN.

1 .....

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2 .....

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[2]

(b) Give **two** characteristics of a LAN.

1 .....

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2 .....

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[2]

(c) One of the company's offices has one server and four computers connected in a star topology.

Draw a diagram to show the layout of the office's star topology.

[2]

(d) Computers can be connected using Ethernet.

Describe what is meant by **Ethernet**.

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..... [3]

(e) The network runs as a thick-client model.

Describe what is meant by a **thick-client** model.

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..... [2]

2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

(a) Identify **two** differences between a WAN and a LAN (local area network).

1 .....

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2 .....

..... [2]

(b) The network uses different topologies in different areas of the sites. In one building there are five computers connected in a mesh topology.

(i) Describe what is meant by a mesh topology.

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..... [2]

(ii) Give **two** advantages of using a mesh topology instead of a bus topology.

1 .....

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2 .....

..... [2]

(c) The computers in one room are set up as thin-clients in a client-server model.

Describe the role of the different computers in this model.

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..... [2]

(d) Students can connect their devices to the university network using cables or a wireless connection.

Explain the benefits to the students of allowing them to use **both** wired and wireless connections.

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..... [4]

(e) One site has split the network into several subnetworks.

An IP address in a subnetwork is divided into two parts.

Identify **and** describe the **two** parts of an IP address in a subnetwork.

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..... [3]

3 A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(a) The company has private cloud storage for its employees to store their work.

(i) Define the term **private cloud**.

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..... [1]

(ii) Describe the benefits to the company of using private cloud storage instead of public cloud storage.

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(b) Part of the internal structure of the wired LAN is a star topology.

Explain how packets are transmitted between two devices in a star topology.

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..... [2]

(c) A different part of the network uses the Ethernet protocol.

(i) A collision is detected.

Describe how the collision is managed using Carrier Sense Multiple Access/Collision Detection (CSMA/CD).

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..... [2]

(ii) Identify **two** drawbacks of using CSMA/CD.

1 .....

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2 .....

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[2]

(d) The devices in the office have static private IP addresses.

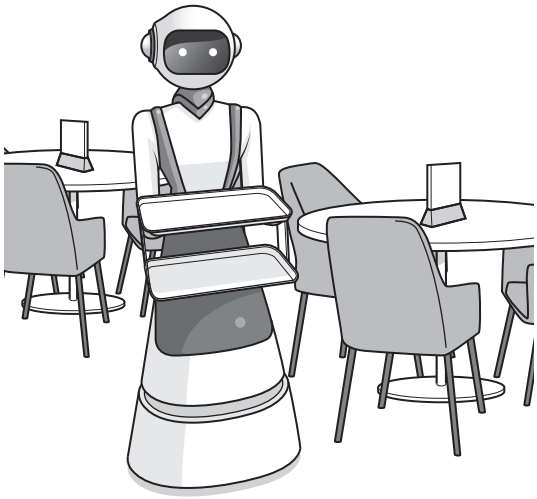
State what is meant by a **static private IP address**.

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..... [1]

4 Robots are used to serve food and drink to customers at a restaurant.



(a) The data from the robots is transmitted to a central computer using a wireless connection.

(i) Complete the table by identifying **and** describing **two** methods of data verification that can be used during data transfer.

	Method	Description
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		.....
2	.....	.....
	.....	.....
		.....
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[4]

(ii) Explain how encryption can protect the security of data during transmission.

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..... [2]

5 A company has multiple sites in different cities.
The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

(a) State **two** ways that a WAN is different to a LAN (Local Area Network).

1 .....

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2 .....

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[2]

(b) The smartphones use the cell phone network to connect to the WAN.

Explain how data is transmitted using the cell phone network.

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..... [4]

(c) Devices that connect to the WAN have an IP address.

Complete the following description of IP addresses by writing the missing words.

IPv4 is displayed as four groups of 8-bit numbers separated by ..... .

Each IPv4 address is 32 bits.

IPv6 is made of eight groups of 4 ..... numbers separated by colons.

Multiple consecutive groups of ..... can be replaced with a double colon.

Each IPv6 address is ..... bits.

A ..... IP address can change each time the computer connects to a network.

A ..... IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.
[6]

(d) The main office for the company has four wired computers, one server, two printers, one central switch and a device providing a single point of access for the internet. The network is set up as a star topology.

Draw the topology for the main office network. Label all devices.

[4]

(e) Describe the role of a switch in a network.

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..... [3]

6 A local area network (LAN) has five computers, one switch and one server.

(a) Describe the characteristics of a LAN.

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(b) Complete the following diagram to show how these devices are connected in a star topology.

Computer Computer Computer Computer










Computer Server Switch

[2]

(c) Describe how packets are transmitted between two hosts using a star topology.

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..... [3]

(d) Another type of network is a bus network.

Ethernet is used to transmit and receive data between the devices on the bus network.

Describe how collisions are detected and managed on this network.

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..... [3]

(e) Complete the table by identifying **one** threat to computer and data security posed by networks and the internet.

Describe the threat and give a method of prevention.

Threat	Description	Prevention
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.....	.....	.....

[3]

END OF QUESTION PAPER